Letters of Recommendation

PhD Application – **Thomas Gravier**

Summary of Contents:

- **Letter 1 – Prof. Gabriel Peyré**
  CNRS Research Director, Département de Mathématiques et Applications, École Normale Supérieure (ENS Ulm)
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  Focus: Academic excellence, theoretical depth, and originality of research work (ICLR 2026 submission).

- **Letter 2 – Dr. Auguste Genovesio**
  Head of Computational Bioimaging and Bioinformatics Lab, IBENS, Inserm Research Director, École Normale Supérieure (ENS Ulm)
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  Focus: Research supervision, scientific rigor, and autonomy during Master's internship.

The following pages contain both letters in full, ordered by author.

(Compiled for submission to the Meta AI PhD program – October 2025.)

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Recommendation letter for Thomas Gravier
PhD fellowship

October 10th, 2025

It is with great enthusiasm that I recommend Thomas Gravier for a PhD fellowship. I had the opportunity to supervise Thomas during his time as a student at the École Normale Supérieure. I also served on the jury for his Master 2 research defense in the highly selective MVA (Mathematics, Vision, Learning) program, widely regarded as the leading French master's program in applied mathematics and artificial intelligence. Throughout his studies and research work, Thomas has demonstrated outstanding intellectual ability, remarkable scientific maturity, and a rare capacity to connect theoretical concepts with concrete applications.

During his research internship at the IBENS laboratory (ENS Ulm), under the supervision of Auguste Genovesio and Thomas Boyer and with my academic guidance, Thomas tackled the challenging problem of probabilistic trajectory modeling from unpaired data. His project, titled *"Multi-Marginal Temporal Schrödinger Bridge Matching for Video Generation from Unpaired Data"*, introduces a new mathematical and algorithmic framework extending Schrödinger Bridge theory to multi-marginal settings. This work bridges rigorous mathematical theory, including optimal transport, stochastic control, and information geometry, with state-of-the-art generative modeling techniques. The project led to a submission to ICLR 2026 (arXiv:2510.01894), a remarkable achievement for a master's thesis and clear evidence of the originality and depth of his research.

Thomas's scientific profile stands out for his mastery of advanced mathematics and his ability to transform theoretical insights into practical, efficient algorithms. His work combined analytical rigor (including proofs of convergence and entropic consistency) with a robust implementation in PyTorch, leveraging GPU-based stochastic simulations. This rare combination of theoretical precision and engineering expertise defines his approach to research: both rigorous and pragmatic, always aiming for reproducibility and clarity in numerical results.

Throughout his internship, Thomas displayed exceptional autonomy, precision, and creativity. He quickly developed a solid understanding of complex mathematical structures such as multi-marginal optimal transport and Schrödinger bridges, and he applied them with confidence to real-world problems such as biological trajectory inference. His presentations were clear, insightful, and accessible to diverse audiences, reflecting both his deep technical understanding and his pedagogical sense.

Thomas graduated from the MVA Master with the highest honors, achieving an exceptional average grade of 17.38/20. His academic excellence, combined with his strong analytical intuition and his enthusiasm for interdisciplinary research, make him an ideal candidate for a

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doctoral program. He has already proven his ability to conduct original and impactful research at the intersection of applied mathematics, machine learning, and probabilistic modeling.

I have no doubt that Thomas Gravier will make substantial contributions to any research group he joins. His intellectual curiosity, rigor, and capacity to connect abstract theory with concrete challenges will be invaluable assets for his doctoral studies and future scientific career. I give him my strongest recommendation for any PhD fellowship or research position he chooses to pursue.

Sincerely,

Gabriel Peyré
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Subject: Recommendation Letter for Thomas Gravier's application to a PhD student program

To the evaluation committee,

It is with great pleasure that I recommend Thomas Gravier, who completed his six-month Master's internship (MVA program) directly supervised by Thomas Boyer in our laboratory on the topic "Dynamical Schrödinger Bridges for video generation from static & unpaired data." This demanding project aimed to extend the theory of Schrödinger Bridges and apply it to the novel setting of video generation from static and unpaired data.

From the very beginning of his internship, Thomas stood out for his remarkable motivation and excellent working capacity. He quickly mastered a subject that is both theoretically complex and technically challenging, requiring a deep understanding of the mathematical foundations of recent Schrödinger Bridge methods as well as strong software engineering skills for their delicate practical implementation. He demonstrated great autonomy and constant initiative, proposing original and relevant ideas to overcome the limitations of existing works.

On the scientific side, Thomas developed a new method adapted to our temporal and multi-marginal framework based on recent advances, and derived solid theoretical results, including a mathematical modeling of our problem and asymptotic convergence proofs for his new algorithm. His rigorous approach and intellectual curiosity allowed him to move fluidly between theory and experimentation, achieving results at the state-of-the-art of the field. He carefully conducted experiments across settings of increasing complexity, from low-dimensional toy examples to high-dimensional image spaces, including comparisons with existing methods on standard benchmarks, demonstrating exemplary experiment management and meticulous attention to detail.

The results of his work have just been submitted as a paper to ICLR, demonstrating excellent scientific quality. This contribution fully reflects his ability to produce high-level research combining originality, rigor, and relevance to the scientific community.

Beyond his technical and scientific skills, Thomas integrated perfectly into the team, actively participating in scientific discussions and collaborating effectively and kindly with his colleagues. His strong engagement within the laboratory and his enthusiasm greatly contributed to the group's positive dynamic.

In summary, Thomas has demonstrated all the qualities expected of a promising young researcher: rigor, creativity, autonomy, and the ability to successfully carry out a complex and ambitious long-term project.

I strongly recommend him for a PhD position, confident that he will continue to show the same level of excellence he displayed during his internship with us. I am convinced that the continuation of this research direction will allow him to fully express his potential.

If you require any further information or have any additional inquiries, please do not hesitate to contact me.

Sincerely,

Auguste Genovesio
October 8th 2025